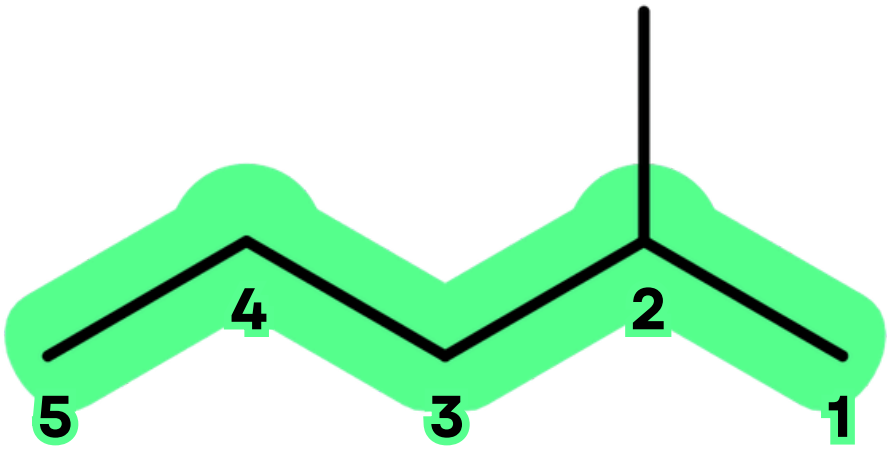
STRUCTURE → NAME

A01 + FOUNDATION · Molecular formula : C_5H_{11}

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: Longest chain = 5 C - ***Pentane.***

Numbering: left to right

Substituents: Methyl on C2 ***2-methyl.***

3 - MY NAME

2-methylpentane

4 - MY ALKANE CHECK



MATCH

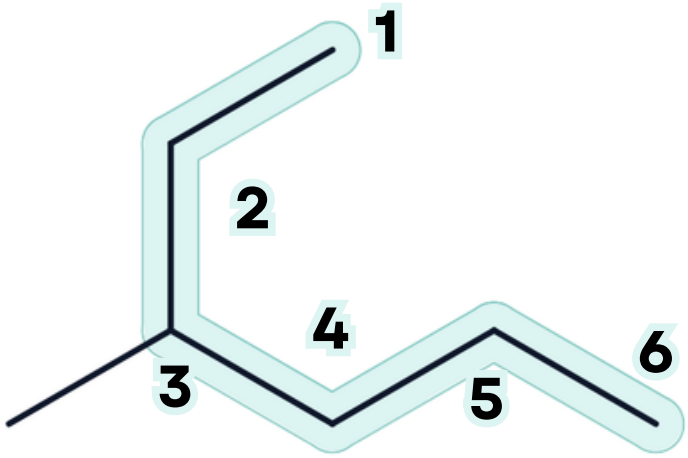
Returned Name : *2-methylpentane*

✓ Longest Chain Confirmed ✓ Lowest Locants ✓ Alphabetical Order ✓ Punctuation ✓ Formula Counted

STRUCTURE → NAME

A02 + FOUNDATION · Molecular formula : C_7H_{16}

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: Longest chain = 6 C - ***Hexane.***

Numbering: right to left

Substituents: Methyl on C3 3-***methyl.***

3 - MY NAME

3-methylhexane

4 - MY ALKANE CHECK



MATCH

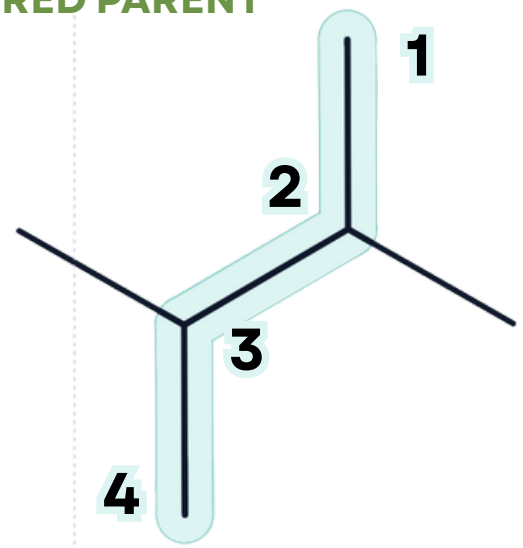
Returned Name : *3-methylhexane*

✓ **Longest Chain Confirmed** ✓ **Lowest Locants** ✓ **Alphabetical Order** ✓ **Punctuation** ✓ **Formula Counted**

STRUCTURE → NAME

A03 + FOUNDATION · Molecular formula : C_6H_{14}

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: Longest chain = 4 C - ***Butane.***

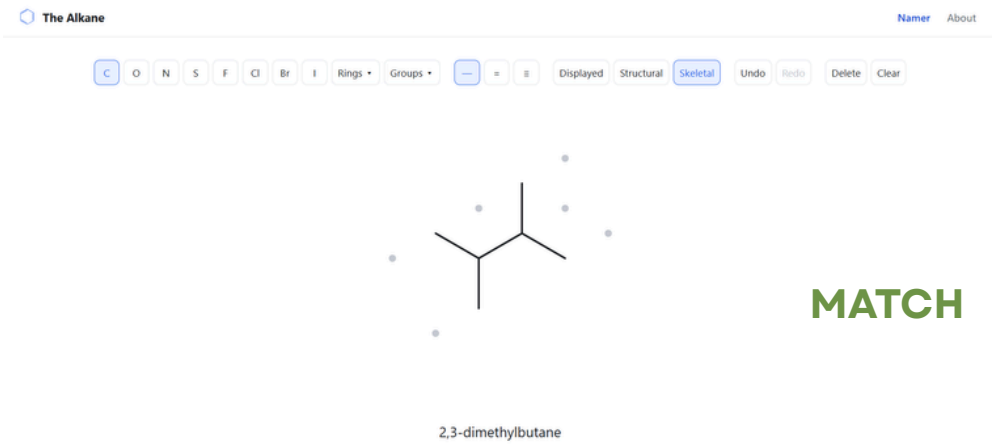
Numbering: right to left

Substituents: Methyl on C2 & C3 ***2,3- dimethyl.***

3 - MY NAME

2,3-dimethylbutane

4 - MY ALKANE CHECK



MATCH

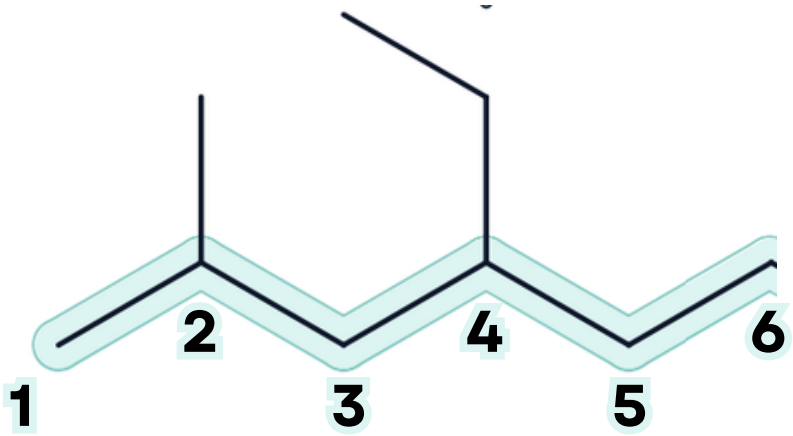
Returned Name : 2,3-dimethylbutane

✓ **Longest Chain Confirmed** ✓ **Lowest Locants** ✓ **Alphabetical Order** ✓ **Punctuation** ✓ **Formula Counted**

STRUCTURE → NAME

A04 + DEVELOPING · Molecular formula : C_9H_{20}

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: Longest chain = 6 C - ***Hexane.***

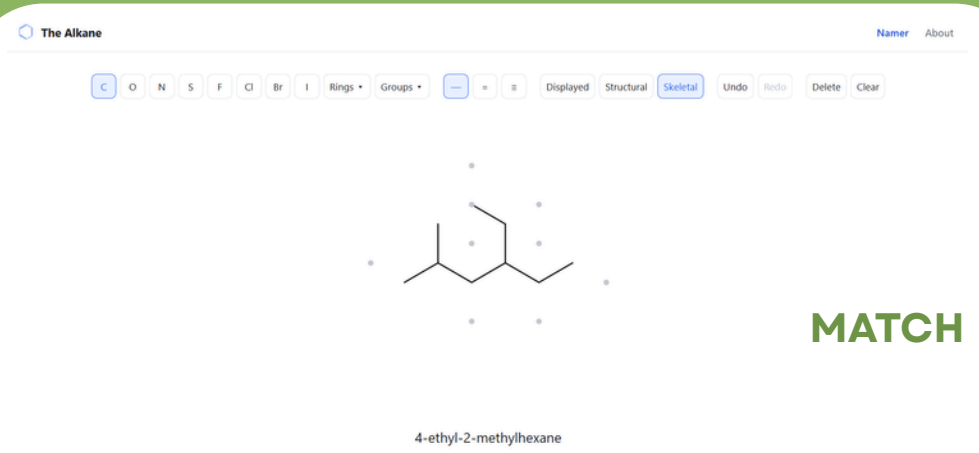
Numbering: Left to Right

Substituents: Methyl on C2 ***2-methyl.***
Ethyl on C4 ***4-ethyl***

3 - MY NAME

4-ethyl-2-methylhexane

4 - MY ALKANE CHECK



MATCH

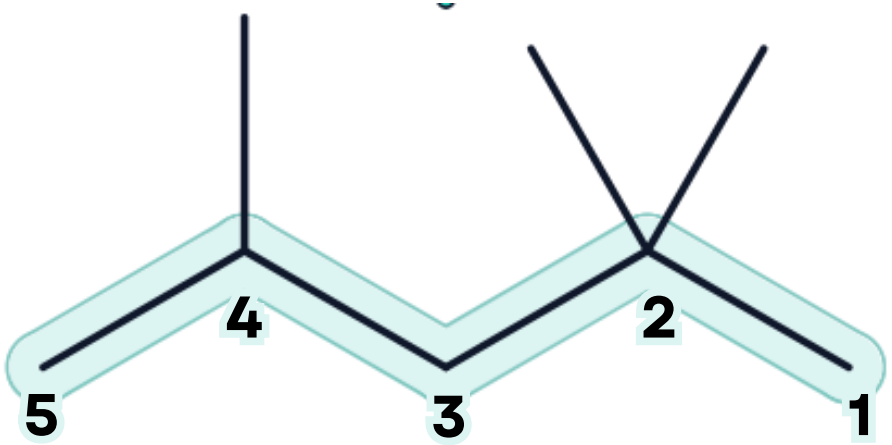
Returned Name : 4-ethyl-2-methylhexane

✓ **Longest Chain Confirmed** ✓ **Lowest Locants** ✓ **Alphabetical Order** ✓ **Punctuation** ✓ **Formula Counted**

STRUCTURE → NAME

A05 + DEVELOPING · Molecular formula : C_8H_{18}

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: Longest chain = 5 C - ***Pentane.***

Numbering: Left to Right

Substituents: 2 Methyl on C2 and One in C4 ***2,2,4-trimethyl.***

3 - MY NAME

2,2,4-trimethylpentane

4 - MY ALKANE CHECK



MATCH

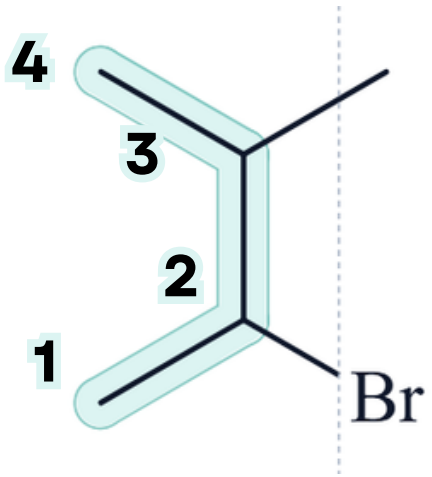
Returned Name : 2,2,4-trimethylpentane

✓ **Longest Chain Confirmed** ✓ **Lowest Locants** ✓ **Alphabetical Order** ✓ **Punctuation** ✓ **Formula Counted**

STRUCTURE → NAME

- *A06 + DEVELOPING · Molecular formula : $C_5H_{11}Br$*

1 - NUMBERED PARENT



2 - NAMING TRAIL

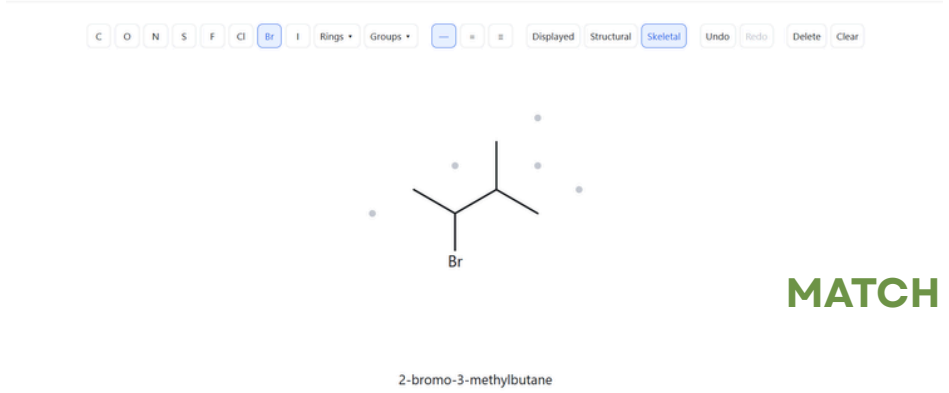
Parent: Longest chain = 4 C - ***Butane.***

Numbering: Left to Right

Substituents: Halogen on C2 & Methyl on C3 ***2-Bromo-3-methyl.***

2-bromo-3-methylbutane

3 - MY NAME



MATCH

Returned Name : 2-bromo-3-methylbutane

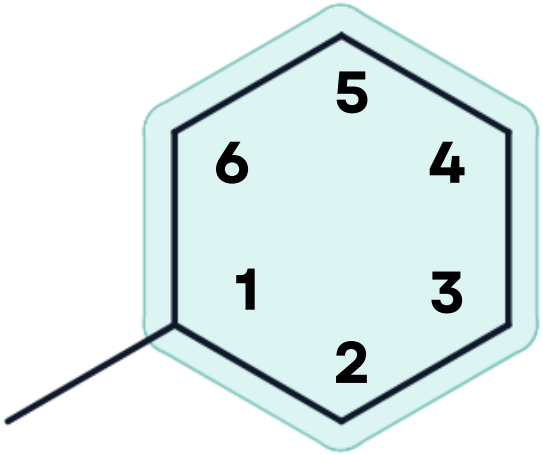
4 - MY ALKANE CHECK

✓ Longest Chain Confirmed ✓ Lowest Locants ✓ Alphabetical Order ✓ Punctuation ✓ Formula Counted

STRUCTURE → NAME

A07 + DEVELOPING · Molecular formula : C_7H_{14}

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: Longest chain = 6 C - cyclo***hexane.***

Numbering: Left to Right

Substituents: Methyl in C1 ***1-methyl.***

3 - MY NAME

1-methylcylohexane

4 - MY ALKANE CHECK



MATCH

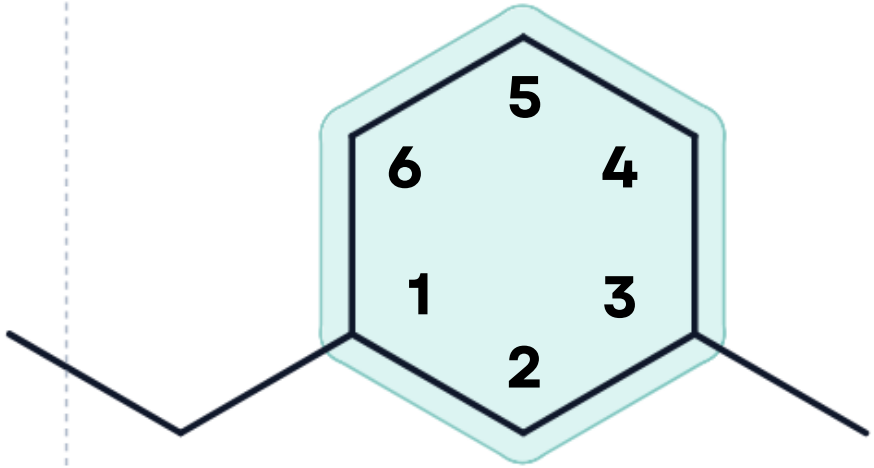
Returned Name : methylcyclohexane

✓ **Longest Chain Confirmed** ✓ **Lowest Locants** ✓ **Alphabetical Order** ✓ **Punctuation** ✓ **Formula Counted**

STRUCTURE → NAME

A08 + CHALLENGE · Molecular formula : C_9H_{18}

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: Longest chain = 6 C - cyclo***hexane.***

Numbering: Left to Right

Substituents: ethyl in C1 & methyl in C2
1-ethyl-3-methyl.

3 - MY NAME

1-ethyl-3-methylcyclohexane

4 - MY ALKANE CHECK



MATCH

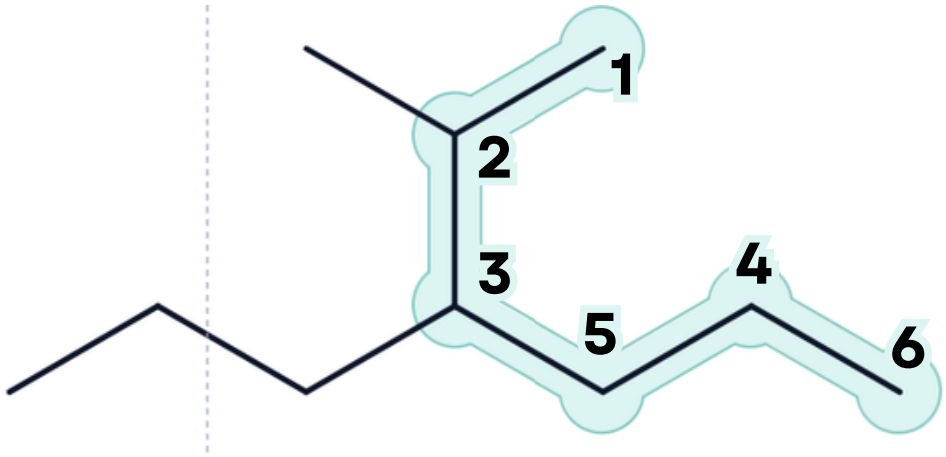
Returned Name : 1-ethyl-3-methylcyclohexane

✓ **Longest Chain Confirmed** ✓ **Lowest Locants** ✓ **Alphabetical Order** ✓ **Punctuation** ✓ **Formula Counted**

STRUCTURE → NAME

A09+ CHALLENGE · Molecular formula : $C_{10}H_{22}$

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: Longest chain = 6 C - ***hexane.***

Numbering: Right to Left

Substituents: Methyl in C2 and Propyl in C3 - ***1-ethyl-3-methyl.***

3 - MY NAME

2-methyl-3-propylhexane

4 - MY ALKANE CHECK



MATCH

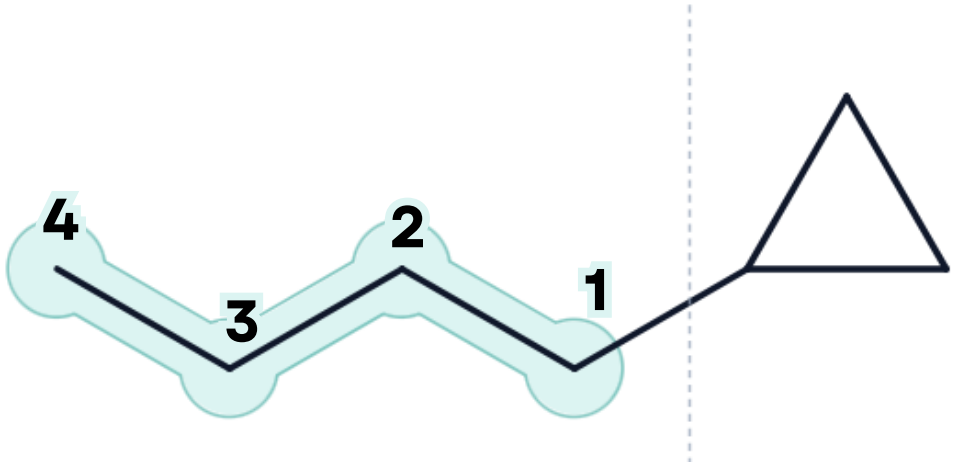
Returned Name : 2-methyl-3-propylhexane

✓ **Longest Chain Confirmed** ✓ **Lowest Locants** ✓ **Alphabetical Order** ✓ **Punctuation** ✓ **Formula Counted**

STRUCTURE → NAME

A10+ CHALLENGE · Molecular formula : C_7H_{14}

1 - NUMBERED PARENT



2 - NAMING TRAIL

Parent: Longest chain = 4 C - ***butane.***

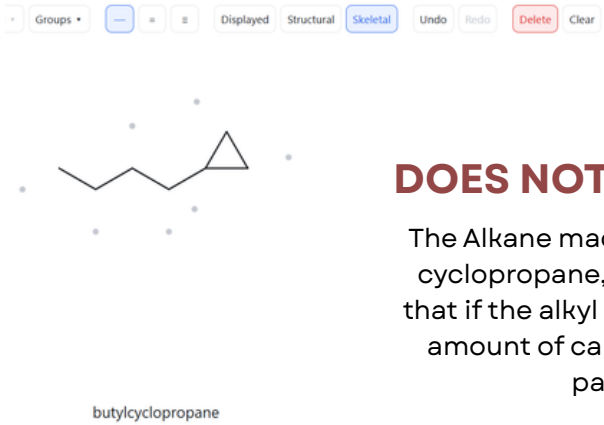
Numbering: Right to Left

Substituents: cyclopropane at C1 - ***1-cyclopropyl***

3 - MY NAME

1-cyclopropylbutane

4 - MY ALKANE CHECK



DOES NOT MATCH

The Alkane made the parent chain the cyclopropane, however IUPAC states that if the alkyl chain posseses a larger amount of carbons it should be the parent chain.

Returned Name : butylcyclopropane

✓ Longest Chain Confirmed ✓ Lowest Locants ✓ Alphabetical Order ✓ Punctuation ✓ Formula Counted

NAME → STRUCTURE

B01 + FOUNDATIONAL · Name: 2-methylpropane

1 - NAME DECODED

Parent: Longest chain = propane

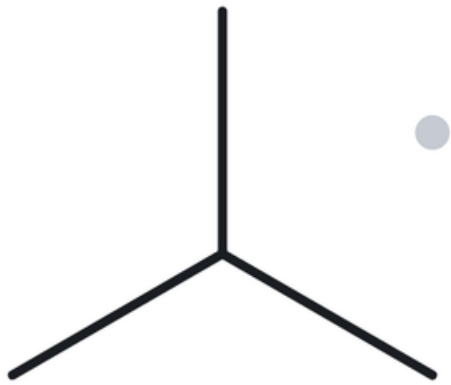
3C

Substituents: 2-methyl

methyl (CH3) in C2

Expected formula: C_4H_{10}

2 - BOND LINE STRUCTURE (THE ALKANE)



Counted from the drawing:

C410

3 - FORMULA CHECK

Returned Name:

2-methylpropane

4 - MY ALKANE CHECK

✓ **Parent length matches** ✓ **Substituent positions match** ✓ **Formula agrees** ✓ **Back-check recorded**

NAME → STRUCTURE

B02+ FOUNDATIONAL · Name: 3-ethylpentane

1 - NAME DECODED

Parent: Longest chain = pentane

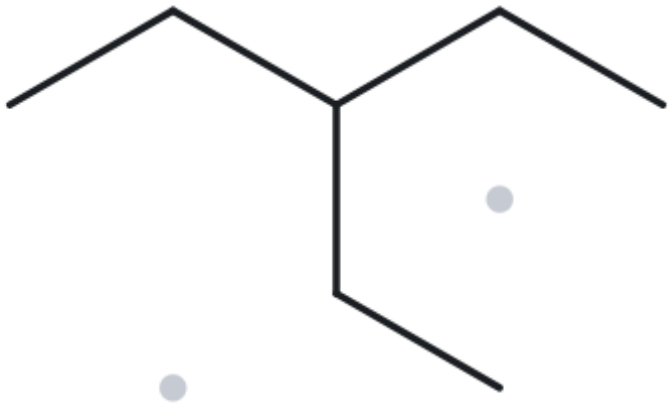
5C

Substituents: 2-ethyl

ethyl (C2H6) in C3

Expected formula: C_7H_{16}

2 - BOND LINE STRUCTURE (THE ALKANE)



Counted from the drawing:

C716

3 - FORMULA CHECK

Returned Name:

3-ethylpentane

4 - MY ALKANE CHECK

✓ **Parent length matches** ✓ **Substituent positions match** ✓ **Formula agrees** ✓ **Back-check recorded**

NAME → STRUCTURE

B03+ FOUNDATIONAL · Name: 2,2-dimethylbutane

1 - NAME DECODED

Parent: Longest chain = butane

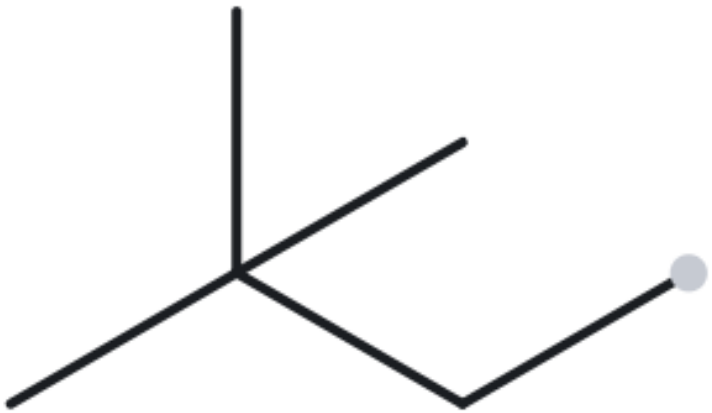
4C

Substituents: 2,2-methyl

2 methyl (CH3) in C2

Expected formula: C_6H_{14}

2 - BOND LINE STRUCTURE (THE ALKANE)



Counted from the drawing:

C614

3 - FORMULA CHECK

Returned Name:

2,2-dimethylbutane

4 - MY ALKANE CHECK

✓ **Parent length matches** ✓ **Substituent positions match** ✓ **Formula agrees** ✓ **Back-check recorded**

NAME → STRUCTURE

B04+ DEVELOPING · Name: 3-ethyl-2-methylpentane

1 - NAME DECODED

Parent: Longest chain = pentane

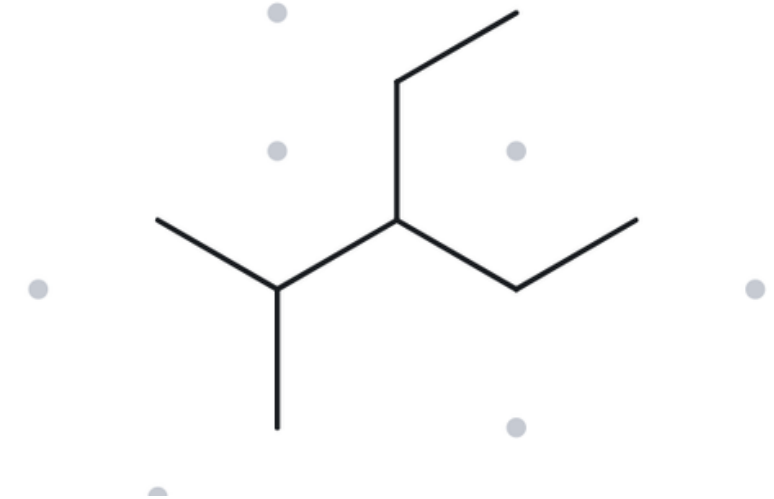
5C

Substituents: 3-ethyl & 2-methyl

1 methyl in C2 & 1 ethyl in C3

Expected formula: C_8H_{18}

2 - BOND LINE STRUCTURE (THE ALKANE)



Counted from the drawing:

C818

3 - FORMULA CHECK

Returned Name:

3-ethyl-2-methylpentane

4 - MY ALKANE CHECK

✓ **Parent length matches** ✓ **Substituent positions match** ✓ **Formula agrees** ✓ **Back-check recorded**

NAME → STRUCTURE

B05+ DEVELOPING · Name: 1-chloro-2-methylpropane

1 - NAME DECODED

Parent: Longest chain = propane

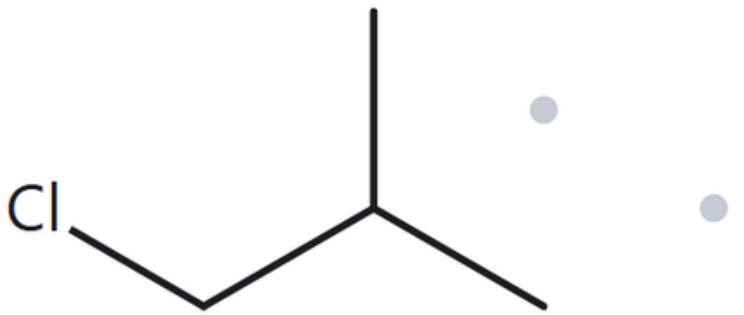
3C

Substituents: 1-chloro & 2-methyl

1 methyl in C2 & 1 chlorine in C1

Expected formula: C_4H_9Cl

2 - BOND LINE STRUCTURE (THE ALKANE)



Counted from the drawing:

C4H9Cl

3 - FORMULA CHECK

Returned Name:

1-chloro-2-methylpropane

4 - MY ALKANE CHECK

✓ Parent length matches ✓ Substituent positions match ✓ Formula agrees ✓ Back-check recorded

NAME → STRUCTURE

B06+ DEVELOPING · Name: 1,2-dimethylcyclopentane

1 - NAME DECODED

Parent: Longest chain = cyclopentane

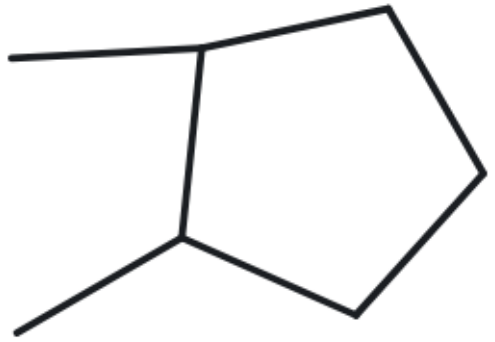
5C

Substituents: 1,2-dimethyl

2 methyl in C1 and C2

Expected formula: C_7H_{14}

2 - BOND LINE STRUCTURE (THE ALKANE)



Counted from the drawing:

C7H14

3 - FORMULA CHECK

Returned Name:

1,2-dimethylcyclopentane

4 - MY ALKANE CHECK

✓ **Parent length matches** ✓ **Substituent positions match** ✓ **Formula agrees** ✓ **Back-check recorded**

NAME → STRUCTURE

B07+ DEVELOPING · Name: 4-ethyl-3,3-dimethylheptane

1 - NAME DECODED

Parent: Longest chain = heptane

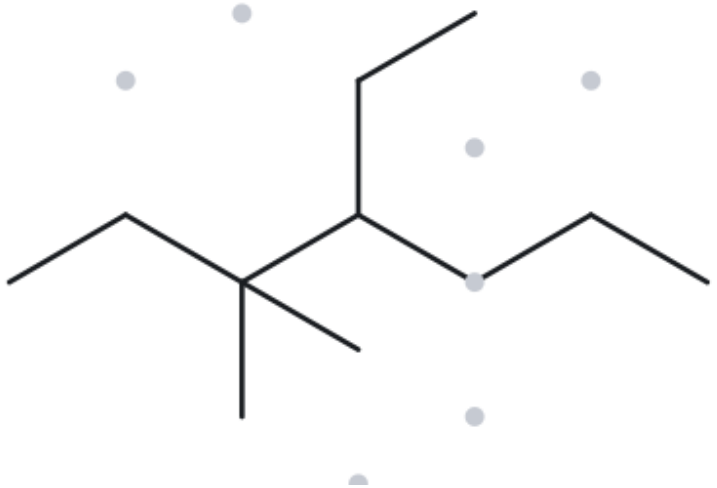
7C

Substituents: 4-ethyl & 3,3-dimethyl

2 methyl in both C3 & 1 ethyl in C4

Expected formula: $C_{11}H_{24}$

2 - BOND LINE STRUCTURE (THE ALKANE)



Counted from the drawing:

C11H24

3 - FORMULA CHECK

Returned Name:

4-ethyl-3,3-dimethylheptane

4 - MY ALKANE CHECK

✓ **Parent length matches** ✓ **Substituent positions match** ✓ **Formula agrees** ✓ **Back-check recorded**

NAME → STRUCTURE

B08+ CHALLENGE · Name: tert-butylcyclohexane

1 - NAME DECODED

Parent: Longest chain = cyclohexane

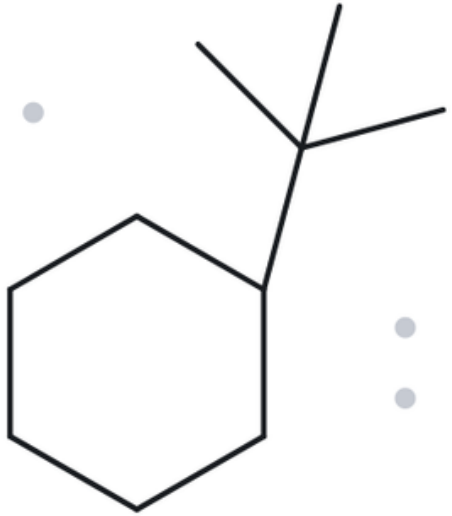
6C

Substituents: tert-butyl

Tert-butyl group in C1

Expected formula: $C_{10}H_{20}$

2 - BOND LINE STRUCTURE (THE ALKANE)



Counted from the drawing:

C10H20

3 - FORMULA CHECK

Returned Name:

(1,1-dimethylethyl)cyclohexane

(The IUPAC name of tert-butyl)

4 - MY ALKANE CHECK

✓ **Parent length matches** ✓ **Substituent positions match** ✓ **Formula agrees** ✓ **Back-check recorded**

NAME → STRUCTURE

B09+ CHALLENGE · Name: 3-ethyl-1,1-dimethylcyclohexane

1 - NAME DECODED

Parent: Longest chain = cyclohexane

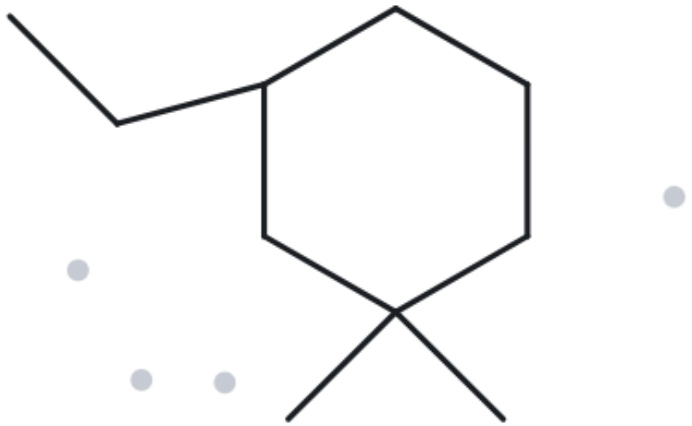
6C

Substituents: 3-ethyl % 1,1-dimethyl

2 methyl both in C1 & 1 ethyl in C3

Expected formula: $C_{10}H_{20}$

2 - BOND LINE STRUCTURE (THE ALKANE)



Counted from the drawing:

C10H20

3 - FORMULA CHECK

Returned Name:

3-ethyl-1,1-dimethylcyclohexane

4 - MY ALKANE CHECK

✓ **Parent length matches** ✓ **Substituent positions match** ✓ **Formula agrees** ✓ **Back-check recorded**

NAME → STRUCTURE

B10 CHALLENGE · Name: 2-ethylbutane

1 - NAME DECODED

Parent: Longest chain = butane

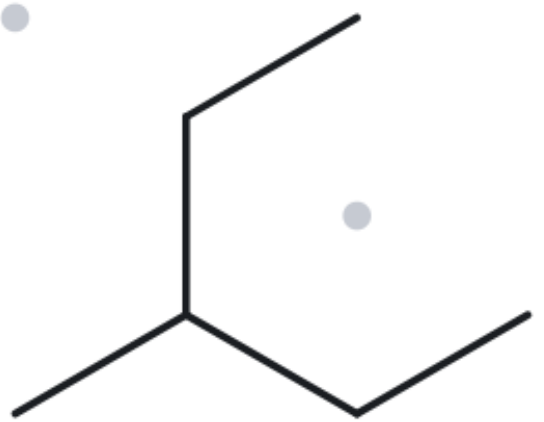
4C

Substituents: 2-ethyl

1 ethyl in C2

Expected formula: C_7H_{16}

2 - BOND LINE STRUCTURE (THE ALKANE)



Counted from the drawing:

C7H16

3 - FORMULA CHECK

Returned Name:

3-methylpentane

4 - MY ALKANE CHECK

✓ **Parent length matches** ✓ **Substituent positions match** ✓ **Formula agrees** ✓ **Back-check recorded**